1 In 2018, the Singapore government announced that personal mobility devices (PMDs) such as e-scooters used on public paths must not have a device speed exceeding 25 km h^{-1} or weigh more than 20 kg.

(a) Suggest why there is a need to set a speed limit and weight limit on PMDs. [2]

Momentum is the product of mass and velocity, which are related to weight and speed of PMD. [1] As force is proportional to rate of change of momentum, the amount of force a PMD can inflict is controlled. [1]
Note: award one mark only answer only state 'Setting speed limit so that rider will not be thrown outward due to sudden stop/deceleration' [1]

(b) Given that a typical e-scooter with rider has a total mass of 100 kg and that the time of impact is 0.20 s, determine the average force this scooter could cause. Assume that the e-scooter comes to rest after impact. [2]

$$F = \frac{\Delta P}{t} = \frac{(100)(0 - \frac{25\times10^3}{60\times60})}{0.20} = 3472 = 3500 \text{ N}$$

(c) Other than the speed and weight of PMDs, state and explain one other factor that can contribute to the magnitude of the force of impact caused in an accident involving PMDs. [2]

- mass of the load on the PMD. A greater load increases the momentum of the PMD/load system, contributing to a large force.

Or

- material of PMD (Hardness of material): Harder material causes shorter time of impact and by N2L, larger force.

(Area of contact is not accepted as it affects the pressure, but not the force)

(d) It is suggested that an infra-red sensor can be fixed at the front end of a PMD such that once an obstacle is detected within 30 cm, the PMD will decelerate to a stop. By reference to the maximum allowable speed of 25 km h^{-1}, explain whether this is feasible. [2]

$$v^2 = u^2 + 2as$$

$$0 = (\frac{25\times10^3}{60\times60})^2 + 2a(0.30)$$

$$a = -80 \text{ m s}^{-2} \approx -8g \qquad [1]$$

Not feasible as the deceleration is several times that of *g*/deceleration is too large. [1]

2 A grenade is launched at ground level with velocity of 75 m s^{-1} and angle 20^0 from the horizontal

For